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Benzimidazole thiophene derivative compounds inducing selective degradation of PLK1

Abstract

The present invention provides novel compounds that induce selective polo-like kinase 1 (PLK1) degradation. Specifically, the present invention provides a bifunctional compound in which a PLK1 binding moiety and an E3 ubiquitin ligase-binding moiety are linked by a chemical linker. The present invention provides the compound, a method for preparing the same, and the use thereof. The compounds may be effectively utilized for preventing or treating PLK1 related diseases.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This is the U.S. National Stage of International Application No. PCT/KR2021/003810, filed Mar. 27, 2021, which was published in English under PCT Article 21(2), which in turn claims the benefit of KR Application No. 10-2020-0037875 on Mar. 27, 2020, which is incorporated herein in its entirety.

TECHNICAL FIELD

(2) The present invention relates to a selective PLK1 degradation inducing compound, a method for

preparing the same, and the use thereof.

BACKGROUND ART

(3) Polo-like kinase 1 (PLK1) is a serine/threonine kinase involved in the conversion of G2/M phase during cell growth and division. PLK1 is expressed and activated in a pulse form from the S phase to the G2/M phase, and rapidly degrades as mitosis ends.

(4) PLK1 is overexpressed in various carcinomas such as colon cancer, lung cancer, bladder cancer, and melanoma, etc., and cancer cells overexpressing PLK1 tend to show resistance to various types of anticancer drugs. As the PLK1 dependence in various carcinomas was revealed as described above, there have been attempts to develop PLK1 inhibitor compounds such as volasertib (also known as BI6727), etc.

(5) However, the conventional PLK1 inhibitors do not sufficiently inhibit PLK1 activity at concentrations that are clinically safe. Thus, there is a problem that even if the cell cycle of cancer cells is temporarily delayed, some cancer cells eventually restart the cell cycle, which may not obtain sufficient clinical effects (see Gheghiani et al., Cell Reports, 2017, etc.). In fact, many pharmaceutical companies such as Boehringer Ingelheim, GlaxoSmithKline, etc., have attempted to develop small-molecular compound-based PLK1 inhibitors, but most of them have failed or stopped in the clinical trial stage, and thus there are no commercially available PLK1 inhibitors to date. It shows that pharmacological mechanism that follows the method of inhibiting enzyme activity by binding to the active site of PLK1 like the small molecule compound inhibitors is not sufficiently effective in the development of new drugs intended to derive anticancer effects by inhibiting PLK1 activity of cancer cells.

(6) Recently, a proteolysis targeting chimera (PROTAC) has been proposed as a small molecule-based platform technology capable of inducing proteolysis of a target protein in the body. The PROTAC is a bifunctional compound in which a ligand molecule that binds to disease-related target protein and an E3 ubiquitin ligase binding moiety are linked by a chemical linker.

Theoretically, the PROTAC compound is capable of inducing degradation of the target protein by placing the disease-related target protein near the E3 ubiquitin ligase.

DISCLOSURE OF INVENTION

Technical Problem

(7) An object of the present invention is to provide selective PLK1 degradation inducing compounds.

(8) Another object of the present invention is to provide a method for preparing the compounds.

(9) Still another object of the present invention is to provide a use of the compounds.

Solution to Problem

(10) Selective PLK1 Degradation Inducing Compounds

(11) The present invention provides novel compounds that induce selective polo-like kinase 1 (PLK1) degradation. Specifically, the present invention provides a bifunctional compound in which a PLK1 binding moiety and an E3 ubiquitin ligase-binding moiety are linked by a chemical linker.

(12) In one general aspect, there is provided a compound represented by the following Formula I, a stereoisomer thereof or a pharmaceutically acceptable salt thereof:

ULM-Linker-PTM [Formula I] in the Formula I above, ULM is E3 ubiquitin ligase binding moiety represented by the following Formula A or B:

(13) ##STR00001## wherein: X.sub.1 is —CH.sub.2— or —CO—;

(14) ##STR00002## wherein:

(15) ##STR00003## is a 5-membered heteroaryl; and Y.sub.1 is hydrogen or C.sub.1-3alkyl;

PTM is PLK1 binding moiety represented by the following Formula II:

(16) ##STR00004##

(17) Linker is a chemical group that links ULM and PTM.

(18) In the Formula A and B,  custom character indicates a covalent bond that links ULM into Linker.

(19) In the Formula II,  custom character indicates a covalent bond that links PTM into Linker.

(20) (1) E3 Ubiquitin Ligase Binding Moiety (ULM)

(21) In one embodiment of the present invention, ULM is a CRBN E3 ubiquitin ligase binding moiety represented by Formula A.

(22) In the present invention, CRBN means Cereblon E3 ubiquitin ligase. CRBN constitutes an E3 ubiquitin ligase complex together with DDB1, Cul4A and ROC1, wherein the CRBN is a substrate recognition subunit of the complex. Some compounds capable of binding to the CRBN E3 ubiquitin ligase are known in the art. For example, after it was known that thalidomide binds to the CRBN E3 ubiquitin ligase (see Ito et al. 2010), it has been reported that a number of immunomodulatory imide drugs (IMiD) including lenalidomide and pomalidomide have CRBN binding ability (see Chamberlain and Brian. 2019; Akuffo et al. 2018; and Burslem et al. 2018, etc.).

(23) In another embodiment of the present invention, ULM is a VHL E3 ubiquitin ligase ligand binding moiety represented by Formula B.

(24) In the present invention, VHL means a von Hippel-Lindau tumor suppressor. VHL constitutes a VCB E3 ligation complex together with Elongin B, Elongin C, CUL2 and Rbx1, wherein VHL is a substrate recognition subunit of the complex. Some compounds capable of binding to the VHL E3 ubiquitin ligase are known in the art. For example, after it was known that peptide such as Ala-Leu-Ala-(Hy)Pro-Tyr-Ile-Pro heptapeptide (see Schneekloth et al. 2004) and Leu-Ala-(Hy)Pro-Tyr-Ile pentapeptide (see Rodriguez-Gonzalez et al. 2008), an improved low-molecular VHL E3 ubiquitin ligase binding compound has been reported (see Buckley et al. J. Am. Chem. Soc. 2012; Buckley et al. Ang. Chem. Int. Ed. 2012; Galdeano et al. 2014; Soares et al. 2017, etc.).

(25) In Formula I of the present invention, the ULM moiety represented by the Formula A or B is covalently linked to the Linker as defined in Formula I through  custom character.

(26) (2) Protein Target Moiety (PTM)

(27) In the compound represented by Formula I, the PTM, a moiety that performs a target protein ligand function, is a polo-like kinase 1 (PLK1) binding moiety represented by Formula II above.

(28) The compound represented by Formula II alone is a benzimidazole thiophene-derived compound that may bind to the active site of PLK1 (see WO2004/014899 and WO2007/030361, etc.)

(29) In one embodiment of Formula I of the present invention, ULM is represented by Formula A wherein Formula II is represented as follows:

(30) ##STR00005##

(31) In one embodiment of Formula I of the present invention, ULM is represented by Formula B wherein Formula II is represented as follows:

(32) ##STR00006##

(33) In a specific embodiment of the present invention, the compound represented by Formula I is a compound that is selected from the group consisting of Compound 1 and 2.

(34) (3) Linker

(35) In one embodiment of the present invention, the Linker as defined in Formula I is represented by the following Formula L:

(36) ##STR00007## wherein:  custom character and  custom character are each independently bond; L.sub.ULM is covalently bonded to ULM moiety through  custom character that is linked thereto, L.sub.PTM is covalently bonded to PTM moiety through  custom character that is linked thereto, L.sub.ULM and L.sub.PTM are each independently a single bond, —CH.sub.2—, —NH—, —O—, —CO—, —CONH— or —NHCO—, L.sub.INT is selected from the group consisting of —CH.sub.2—, —CH.sub.2CH.sub.2—, —CHCH—, —CC—, —CH

(37) ##STR00008## {wherein

(38) ##STR00009## is 3-10 membered cycloalkyl, 4-10 membered heterocycloalkyl, 6-10 membered aryl, or 5-10 membered heteroaryl}; and p is an integer between 1 to 10.

- (39) In one embodiment, Linker is a linker that is included in the compound selected from the group consisting of Compound 1 and 2.
- (40) In a certain embodiment of the present invention, the compound represented by Formula I is a compound that is selected from the group consisting of Compound 1 and 2.
- (41) In the present invention, a pharmaceutically acceptable salt refers to any organic or inorganic acid addition salt with a concentration that is relatively non-toxic, is harmless, and has effective action to patients, wherein side effects caused by this salt does not deteriorate beneficial efficacy of the compound represented by Formula I. For example, the pharmaceutically acceptable salt may be an inorganic acid such as hydrochloric acid, phosphoric acid, sulfuric acid, nitric acid, or the like, or an organic acid such as methanesulfonic acid, p-toluenesulfonic acid, acetic acid, trifluoroacetic acid, maleic acid, succinic acid, oxalic acid, benzoic acid, tartaric acid, fumaric acid, mandelic acid, propionic acid, citric acid, lactic acid, glycolic acid, gluconic acid, galacturonic acid, glutamic acid, glutaric acid, glucuronic acid, aspartic acid, ascorbic acid, carbonic acid, vanillic acid or hydroiodic acid, but is not limited thereto.
- (42) Method for the Preparing the Selective PLK1 Degradation Inducing Compounds
- (43) In the present invention, the compound represented by Formula I above, a stereoisomer thereof, or a pharmaceutically acceptable salt thereof may be prepared through reactions such as the following Reaction Schemes 1 to 3 by a synthetic method known in the field of organic chemistry or a modification technique apparent to those skilled in the art.
- (44) ##STR00010##
- (45) ##STR00011##
- (46) ##STR00012##
- (47) In the Reaction Schemes 1 to 3 above, PTM, Linker and ULM are a group defined in the above, or a suitable derivative thereof. RG.sup.1, RG.sup.2, RG.sup.2a, RG.sup.2b, RG.sup.3, RG.sup.3a, RG.sup.3b and RG.sup.4 are moieties including a suitable reactive group capable of linking together with an intermediate of the PROTAC compound represented by Formula I through formation of the covalent bond in the field of organic synthesis. The formation of the covalent bond may be achieved by synthetic reactions such as amide formation, ester formation, carbamate formation, urea formation, ether formation, amine formation, and single bonds, double bond formation between various carbons, click chemistry and the like, depending on specific reaction groups, but is not limited thereto.
- (48) Variations of each step in the above Reaction Scheme may include one or multiple synthesis steps. Isolation and purification of the product may be accomplished by standard procedures known to those skilled in the art of organic chemistry.
- (49) In the present invention, a preparation method of Example 1 is presented as an example of Reaction Scheme 1, and a preparation method of Example 2 is presented as an example of Reaction Scheme 3, but the preparation methods of the present invention are not limited thereto.
- (50) In the above Reaction Scheme, each compound represented by PTM and ULM may be synthesized by a person skilled in the art with reference to documents known in the field of organic chemistry, descriptions of Examples of the present invention, and the like.
- (51) The present invention also provides the compounds represented by PTM-Linker-RG.sup.3 or PTM-Linker 1-RG.sup.2b that are the reaction intermediates of the compounds represented by Formula I.
- (52) Use of the Selective PLK1 Degradation Inducing Compounds
- (53) An embodiment of the present invention is a composition for inducing PLK1 degradation including a compound represented by Formula I or a pharmaceutically acceptable salt thereof. The Formula I is the same as defined above.
- (54) The PLK1 degradation-inducing PROTAC compound of the present invention is capable of fundamentally degrading the target protein, PLK1 in view of the mechanism of action, thereby achieving an excellent PLK1 inhibitory effect as compared to the conventional PLK1 small

molecule inhibitor that inhibits the simple activity of PLK1.

(55) Accordingly, the composition including the compound represented by Formula I of the present invention or a pharmaceutically acceptable salt thereof may be effectively employed for selective degradation of PLK1.

(56) An embodiment of the present invention is a composition for preventing or treating PLK1-related diseases including the compound represented by Formula I or the pharmaceutically acceptable salt thereof. Another embodiment of the present invention is a method for the prevention or treatment of PLK1-related diseases comprising administering the composition to a subject in need thereof. The Formula I is the same as defined above.

(57) In the present invention, the PLK1-related disease refers to any disease or condition capable of being treated, alleviated, delayed, inhibited or prevented from induction of degradation or inhibition of activity of PLK1. In an embodiment, the PLK1-related disease may be a cancer (malignant tumor), a benign tumor, a neurological disease, or other genetic or non-genetic diseases caused by excessive cell division.

(58) The cancer includes all cancers capable of exhibiting prophylactic or therapeutic efficacy due to inhibition of PLK1 activity, and may be solid cancer or blood cancer. For example, the cancer may be one or more selected from the group consisting of squamous cell carcinoma, small cell lung cancer, non-small cell lung cancer, lung adenocarcinoma, lung squamous cell carcinoma, peritoneal cancer, skin cancer, skin or intraocular melanoma, rectal cancer, anal muscle cancer, esophageal cancer, small intestine cancer, endocrine cancer, parathyroid cancer, adrenal cancer, soft tissue sarcoma, urethral cancer, chronic or acute leukemia, lymphocytic lymphoma, hepatocellular carcinoma, gastrointestinal cancer, gastric cancer, pancreatic cancer, glioblastoma, cervical cancer, ovarian cancer, liver cancer, bladder cancer, liver tumor, breast cancer, colon cancer, colorectal cancer, endometrial or uterine cancer, salivary gland cancer, kidney cancer, prostate cancer, vulvar cancer, thyroid cancer, head and neck cancer, brain cancer, osteosarcoma, and the like, but is not limited thereto. The cancer includes not only primary cancer but also metastatic cancer.

(59) The benign tumors include all benign tumors capable of exhibiting prophylactic or therapeutic efficacy due to the inhibition of PLK1 activity, such as benign tumors in pre-cancer stages, and may be solid tumors or blood tumors. For example, the tumor may be one or more selected from the group consisting of Barrett's esophagus, colon adenoma and polyp, breast fibroadenoma and cyst, monoclonal gammopathy of undetermined significance (MGUS), monoclonal lymphocytosis, and the like, but is not limited thereto.

(60) The neurological diseases include all neurological diseases capable of exhibiting prophylactic or therapeutic efficacy due to the inhibition of PLK1 activity, and specifically, may be one or more selected from the group consisting of central nervous system disease, neurodegenerative disease, Alzheimer's disease, Parkinson's disease, multiple sclerosis, Huntington's disease, senile dementia, epilepsy, Lou Gehrig, stroke, and nerve damage and axonal degeneration-related disorders following brain or spinal cord injury, but is not limited thereto.

(61) The pharmaceutical composition of the present invention may further include one or more active ingredients exhibiting the same or similar medicinal effects in addition to the compound represented by Formula I above, or the pharmaceutically acceptable salt thereof.

(62) An embodiment of the present invention is a method of degrading PLK1 by administering a compound represented by Formula I or a pharmaceutically acceptable salt thereof to mammals including humans.

(63) Another embodiment of the present invention is a method of degrading PLK1 by administering the compound represented by Formula I or the pharmaceutically acceptable salt thereof to a sample in vitro. The sample may be a cell, a cell culture, a body fluid or tissue of a mammal including a human, but is not limited thereto.

Advantageous Effects of Invention

(64) The compound of the present invention exhibits an effect of inducing PLK1 degradation.

Therefore, the pharmaceutical compound of the present invention may be effectively utilized for preventing or treating PLK1-related diseases.

Description

BEST MODE FOR CARRYING OUT THE INVENTION

(1) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. The terminology used in the description is for describing particular embodiments only and is not intended to be limiting of the disclosure.

(2) The present invention provides synthetic methods for compounds 1 and 2 shown in the table below.

(3) TABLE-US-00001 TABLE 1 Compound Structure 1  embedded image 2  embedded image

(4) The compounds of the present invention were purified according to the following method and the structure was analyzed.

(5) Instruments

(6) LCMS: Shimadzu LCMS-2020 NMR: BRUKER AVANCE III 400 MHz HPLC: Shimadzu LC-20AB, Shimadzu LC-20AD, Agilent 1100 LC, Agilent 1200 LC, Agilent 1290 LC

LCSM Analysis

(7) LCMS data were recorded with Shimadzu LCMS-2020 equipped with an electron spray ionization device. 0.0375% TFA in water (solvent A) and 0.01875% TFA in acetonitrile (solvent B) were used as mobile phases. As a column, Kinetex EVO C18 (2.1*30) mm, 5 um was used.

(8) HPLC Analysis

(9) In HPLC analysis, Shimadzu LC-20AB, Shimadzu LC-20AD, Agilent 1100 LC, Agilent 1200 LC or Agilent 1290 LC was used. 0.0375% TFA in water (solvent A) and 0.01875% TFA in acetonitrile (solvent B) or 0.025% NH₃·H₂O in water (solvent A) and acetonitrile (Solvent B) was used as the mobile phase. As a column, XBridge C18 (2.1*50) mm, 5 um or Kinetex C18 LC column (4.6*50) mm, 5 um or Eclipse plus C18 (4.6*150) mm, 3.5 um or Waters XBridge® C18 (4.6*150) mm, 3.5 μm was used.

(10) NMR Analysis

(11) ¹H NMR spectrum was recorded with Bruker AVANCE III 400 MHz/5 mm Probe (BBO).

Example 1. Synthesis of 3-(1-(4-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)-2-(trifluoromethyl)phenyl)ethoxy)-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophene-2-carboxamide (Compound 1)

(12) ##STR00015## ##STR00016## ##STR00017## ##STR00018##

Step 1: Synthesis of methyl 4-hydroxy-2-(trifluoromethyl)benzoate (2)

(13) To a solution of 4-hydroxy-2-(trifluoromethyl)benzoic acid (7.5 g, 36.39 mmol) in toluene (75 mL) and MeOH (37 mL) was added TMSCHN₂ (2 M, 36.4 mL) and the mixture was stirred at 25° C. for 2 h. TLC (Petroleum ether:Ethyl acetate=2:1) showed the starting material was consumed and new spot was detected. The mixture was concentrated under reduced pressure to afford methyl 4-hydroxy-2-(trifluoromethyl)benzoate (8 g, crude) was obtained as yellow solid.

Step 2: Synthesis of 3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl 4-methylbenzenesulfonate (3)

(14) To a solution benzyl (2-(2-(2-hydroxyethoxy)ethoxy)ethyl)carbamate (6.9 g, 24.35 mmol), TEA (7.39 g, 73.06 mmol, 10.17 mL) in DCM (50 mL) was added TosCl (9.29 g, 48.71 mmol) at 25° C. The solution was stirred at 25° C. for 12 h. LCMS showed a main peak with desired mass. The mixture was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (45 g SepaFlash® Silica Flash Column, Eluent of 0~65% Ethylacetate/Petroleum ether gradient @ 75 mL/min) to afford 3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl 4-

methylbenzenesulfonate (9.1 g, 20.59 mmol, 84.55% yield, 99% purity) as colorless oil.

MS(M+H).sup.+ = 438.0.

Step 3: Synthesis of methyl 4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)benzoate (4)

(15) To a solution of 3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl 4-methylbenzenesulfonate (8.9 g, 20.34 mmol) and methyl 4-hydroxy-2-(trifluoromethyl)benzoate (4.93 g, 22.38 mmol) in ACN (180 mL) was added K.sub.2CO.sub.3 (5.62 g, 40.69 mmol) and KI (168.84 mg, 1.02 mmol) and the mixture was stirred at 80° C. for 14 h. LCMS showed the desired mass was detected. The mixture was filtered and the filter cake was washed with EtOAc (100 mL). The filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (20 g SepaFlash® Silica Flash Column, Eluent of 0~30% Ethyl acetate/Petroleum ether gradient @ 60 mL/min) to afford methyl 4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)benzoate (7.4 g, 14.94 mmol, 73.44% yield, 98% purity) as yellow oil.

MS(M+H).sup.+ = 486.2.

Step 4: Synthesis of benzyl (2-(2-(2-(4-acetyl-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (5)

(16) To a solution of MeMgBr (3 M, 35.50 mL) in toluene (49 mL) was added TEA (7.74 g, 76.52 mmol, 10.65 mL) at 0° C. and a solution of methyl 4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)benzoate (7.1 g, 14.63 mmol) in toluene (49 mL) was added and the resulting mixture was stirred at 25° C. for 14 h. LCMS showed the desired mass was detected. The mixture was quenched with NH.sub.4Cl (10 mL) at 0° C. and extracted with EtOAc (10 mL×3). The combined organic layers were washed with H.sub.2O (10 mL) and dried over Na.sub.2SO.sub.4, filtered. The filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (40 g SepaFlash® Silica Flash Column, Eluent of 10~60% Ethyl acetate/Petroleum ether gradient @ 100 mL/min) to afford benzyl (2-(2-(2-(4-acetyl-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (5.4 g, 10.93 mmol, 74.72% yield, 95% purity) as yellow oil. MS(M+H).sup.+ = 470.2

Step 5: Synthesis of benzyl (2-(2-(2-(4-(1-hydroxyethyl)-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (6)

(17) To a solution of benzyl (2-(2-(2-(4-acetyl-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (1.8 g, 3.83 mmol) in MeOH (40 mL) was added NaBH₄ (450 mg, 11.90 mmol) slowly and the mixture was stirred at 25° C. for 1 h. LCMS showed the desired mass was detected. The mixture was quenched with NH.sub.4Cl (30 mL) and concentrated under reduced pressure. The mixture was extracted with EtOAc (20 mL×3). The combined organic layers were washed with NaHCO.sub.3 (10 mL×2) and H.sub.2O (10 mL×2), dried over Na.sub.2SO.sub.4, filtered. The filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (12 g SepaFlash® Silica Flash Column, Eluent of 20~80% Ethyl acetate/Petroleum ether gradient @ 50 mL/min) to afford benzyl (2-(2-(2-(4-(1-hydroxyethyl)-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (1.77 g, 3.75 mmol, 97.91% yield, 100% purity) as yellow oil. MS(M+Na).sup.+ = 494.2

Step 6: Synthesis of methyl 5-nitro-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (8)

(18) To a solution of benzyl (2-(2-(2-(4-(1-hydroxyethyl)-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (1.77 g, 3.75 mmol), methyl 3-hydroxy-5-nitrothiophene-2-carboxylate (915 mg, 4.50 mmol) and PPh.sub.3 (1.48 g, 5.63 mmol) in DCM (40 mL) was added DTBAD (1.30 g, 5.63 mmol) at 25° C. The mixture was stirred at 25° C. for 14 h. The mixture was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (20 g SepaFlash® Silica Flash Column, Eluent of 10-100% Ethyl acetate/Petroleum ether gradient @ 50 mL/min). The crude product was triturated with MTBE (10 mL) and filtered. The filter cake was collected and dried under reduced pressure to afford methyl 5-

nitro-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (2.06 g, 3.14 mmol, 83.57% yield) as yellow oil.

Step 7: Synthesis of methyl 5-amino-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (9)

(19) To a solution of methyl 5-nitro-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (2.06 g, 3.14 mmol) in EtOH (20 mL) and H₂O (20 mL) was added NH₄Cl (1.01 g, 18.82 mmol) and Fe (1.05 g, 18.82 mmol) and the resulting mixture was stirred at 80° C. for 1 h. LCMS showed the desired mass was detected. The mixture was filtered at 80° C. and the filter cake was washed with MeOH (50 mL). The filtrate was concentrated under reduced pressure to remove MeOH and extracted with EtOAc (20 mL×3). The combined organic layer was washed with H₂O (10 mL×2), dried over Na₂SO₄, filtered. The filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (20 g SepaFlash® Silica Flash Column, Eluent of 20~60% Ethyl acetate/Petroleum ether gradient @ 50 mL/min) to afford methyl 5-amino-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (0.9 g, 1.44 mmol, 45.78% yield) as yellow oil. MS(M+H).sup.+ = 627.4

Step 8: Synthesis of 1-(3-bromo-4-nitrobenzyl)-4-methylpiperazine (10)

(20) To a solution of 1-methylpiperazine (866.88 mg, 8.65 mmol, 960 µL) and 3-bromo-4-nitrobenzaldehyde (1 g, 4.35 mmol) in DCE (30 mL) was added AcOH (10.50 mg, 174.85 µmol, 10 µL) at 0° C. and the mixture was stirred at 0° C. for 30 min, then NaBH(OAc) 3 (2.77 g, 13.05 mmol) was added and the resulting mixture was stirred at 25° C. for 14 h. LCMS showed the desired mass was detected. The mixture was diluted with H₂O (20 mL) and citric acid solution (50 mL), then washed with EtOAc (10 mL×2), the organic layer was discarded. The aqueous phase was adjusted the pH=9 and extracted with EtOAc (10 mL×2). The combined organic layers was dried over Na₂SO₄, filtered. The filtrate was concentrated under reduced pressure to afford 1-(3-bromo-4-nitrobenzyl)-4-methylpiperazine (1.2 g, 3.74 mmol, 86.10% yield, 98% purity) as yellow oil. MS(M+H).sup.+ = 314.1

Step 9: Synthesis of methyl 5-((5-((4-methylpiperazin-1-yl)methyl)-2-nitrophenyl)amino)-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (11)

(21) To a solution of 1-(3-bromo-4-nitrobenzyl)-4-methylpiperazine (355.14 mg, 1.13 mmol) and methyl 5-amino-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (850.00 mg, 1.36 mmol) in toluene (15 mL) were added Cs₂CO₃ (1.10 g, 3.39 mmol), Pd₂(dba)₃ (20.70 mg, 22.61 µmol) and Xantphos (28.78 mg, 49.74 µmol) and the resulting mixture was stirred at 80° C. for 14 h. TLC (Ethyl acetate:Methanol=5:1) showed the desired spot was detected. The mixture was filtered and the filter cake was washed with EtOAc (50 mL). The filtrate was concentrated under reduced pressure. Then the crude product was diluted with H₂O (10 mL) and extracted with EtOAc (10 mL×2). The combined organic was washed with H₂O (10 mL×2), dried over Na₂SO₄, filtered. The filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (5 g SepaFlash® Silica Flash Column, Eluent of 50-100% Ethyl acetate/Petroleum ether to 0~20% Methanol/Ethyl acetate gradient @ 50 mL/min) to afford methyl 5-((5-((4-methylpiperazin-1-yl)methyl)-2-nitrophenyl)amino)-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (510 mg, 593.09 µmol, 52.47% yield) as yellow oil. MS(M+H).sup.+ = 860.4

Step 10: Synthesis of methyl 5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (12)

(22) To a solution of methyl 5-((5-((4-methylpiperazin-1-yl)methyl)-2-nitrophenyl)amino)-3-(1-(4-

((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (410 mg, 476.80 μmol) in MeOH (8 mL) was added trimethoxymethane (1.27 g, 11.97 mmol, 1.31 mL), formic acid (200.09 mg, 4.35 mmol, 164.00 μL) and Zn (156 mg, 2.39 mmol). The mixture was stirred at 70° C. for 6 h. LCMS showed the desired mass was detected. The mixture was filtered and the filter cake was washed with hot MeOH (50 mL, 50° C.) and the filtrate was concentrated under reduced pressure. The crude was diluted with H.sub.2O (10 mL) and adjusted the pH=8 with NaHCO.sub.3 solution. Then the mixture was extracted with EtOAc (10 mL $\times$ 3). The combined organic layers was washed with H.sub.2O (10 mL $\times$ 2), dried over Na.sub.2SO.sub.4, filtered. The filtrate was concentrated under reduced pressure to afford methyl 5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (330 mg, crude) as yellow oil.
MS(M+H).sup.+ = 840.5

Step 11: Synthesis of 5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylic acid (13)
(23) To a solution of methyl 5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (250.00 mg, 297.65 μmol) in THF (3 mL) was added a solution of LiOH.custom character.sub.2O (14.39 mg, 342.89 μmol) in H.sub.2O (1.5 mL) and the mixture was stirred at 40° C. for 14 h. Additional LiOH.custom character.sub.2O (17.98 mg, 428.61 μmol) was added and the mixture was stirred at 40° C. for another 14 h. LCMS showed the starting material was consumed and the desired mass was detected. The mixture was concentrated under reduced pressure to afford 5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylic acid (250 mg, crude) as yellow oil.
MS(M+H).sup.+ = 826.6

Step 12: Synthesis of benzyl (2-(2-(2-(4-(1-((2-carbamoyl-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophen-3-yl)oxy)ethyl)-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (14)
(24) To a solution of 5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(4-((3-oxo-1-phenyl-2,7,10-trioxa-4-azadodecan-12-yl)oxy)-2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylic acid (250 mg, 300.54 μmol) and NH.sub.4Cl (80.38 mg, 1.50 mmol) in DMF (4 mL) were added DIPEA (118.72 mg, 918.58 μmol , 160 μL) and HATU (171.41 mg, 450.82 μmol) and the resulting mixture was stirred at 25° C. for 14 h. LCMS showed the starting material remained. Additional NH.sub.4Cl (80.38 mg, 1.50 mmol) and HATU (171.41 mg, 450.82 μmol) were added and the resulting mixture was stirred at 25° C. for 1 h. LCMS showed the desired mass was detected. The reaction mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (10 mL $\times$ 3). The combined organic layer was washed with brine (10 mL $\times$ 2), dried over Na.sub.2SO 4, filtered. The filtrate was concentrated under reduced pressure to afford benzyl (2-(2-(2-(4-(1-((2-carbamoyl-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophen-3-yl)oxy)ethyl)-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (170 mg, crude) as yellow oil.
MS(M+H).sup.+ = 825.6

Step 13: Synthesis of 3-(1-(4-(2-(2-(2-aminoethoxy)ethoxy)ethoxy)-2-(trifluoromethyl)phenyl)ethoxy)-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophene-2-carboxamide (15)
(25) To a solution of benzyl (2-(2-(2-(4-(1-((2-carbamoyl-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophen-3-yl)oxy)ethyl)-3-(trifluoromethyl)phenoxy)ethoxy)ethoxy)ethyl)carbamate (150 mg, 181.84 μmol) in ACN (0.75

mL) was added TMSI (220.50 mg, 1.10 mmol, 150 μ L) and the mixture was stirred at 20° C. for 1 h. LCMS showed the starting material was consumed and the desired mass was detected. The mixture was quenched with H.sub.2O (20 mL, 0° C.) and washed with EtOAc (10 mL $\times$ 3), the organic layer was discarded. The aqueous phase was adjusted the pH=7-8 with Na.sub.2CO.sub.3 solution, then the mixture was lyophilized. The crude product was purified by prep-HPLC (column: Unisil 3~100 C.sub.18 μ Ltra 150*50 mm*3 μ m; mobile phase: [water (0.225% FA)-ACN]; B %: 5%-35%, 10 min) to afford 3-(1-(4-(2-(2-(2-aminoethoxy)ethoxy)ethoxy)-2-(trifluoromethyl)phenyl)ethoxy)-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophene-2-carboxamide (60 mg, 70.22 μ mol, 38.62% yield, 97% purity, FA salt) as yellow solid. MS(M+H).sup.+ = 691.4

Step 14: Synthesis of 3-(1-(4-(2-(2-(2-((2-(2,6 dioxopiperidin-3-yl)-1,3 dioxoisindolin-4-yl) amino)ethoxy)ethoxy)ethoxy)-2-(trifluoromethyl)phenyl)ethoxy)-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophene-2-carboxamide (Compound 1)

(26) To a solution of 3-(1-(4-(2-(2-(2-aminoethoxy)ethoxy)ethoxy)-2-(trifluoromethyl)phenyl)ethoxy)-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophene-2-carboxamide (60 mg, 72.39 μ mol, FA salt) and 2-(2,6-dioxopiperidin-3-yl)-4-fluoroisindoline-1,3-dione (24 mg, 86.89 μ mol) in DMSO (1.2 mL) was added TEA (43.95 mg, 434.34 μ mol, 60.45 μ L) and the mixture was stirred at 80° C. for 14 h. LCMS showed the desired mass was detected. The mixture was filtered and the filtrate was purified by prep-HPLC (column: Phenomenex Gemini-NX C.sub.18 75*30 mm*3 μ m; mobile phase: [water (10 mM NH.sub.4HCO₃)-ACN]; B %: 30%-60%, 8 min) and the eluent was lyophilized to afford 3-(1-(4-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1, 3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy) ethoxy)-2-(trifluoromethyl)phenyl)ethoxy)-5-(6-((4-methylpiperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)thiophene-2-carboxamide (10.2 mg, 9.80 μ mol, 13.54% yield, 91% purity) as yellow solid. MS(M+H).sup.+ = 947.7

(27) .sup.1H NMR (400 MHz, CD.sub.3CN) δ =8.06 (d, J=2.5 Hz, 1H), 7.69 (d, J=8.7 Hz, 1H), 7.64 (d, J=8.3 Hz, 1H), 7.51~7.44 (m, 1H), 7.33 (d, J=5.6 Hz, 1H), 7.30-7.27 (m, 1H), 7.24-7.17 (m, 2H), 7.11 (br s, 1H), 7.0-6.94 (m, 2H), 6.85 (d, J=1.0 Hz, 1H), 6.48-6.43 (m, 1H), 6.29 (br s, 1H), 5.88-5.82 (m, 1H), 4.91-4.86 (m, 1H), 4.14-4.09 (m, 2H), 3.78-3.74 (m, 2H), 3.66-3.57 (m, 6H), 3.56-3.52 (m, 2H), 3.42-3.37 (m, 2H), 2.74-2.61 (m, 4H), 2.45-2.30 (m, 8H), 2.06-2.00 (m, 2H), 1.78-1.71 (m, 4H).

Example 2. Synthesis of (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-((1-(5-carbamoyl-4-((R)-1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazin-1-yl)-4-oxo-6,9,12-trioxa-3-azatetradecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Compound 2)

(28) ##STR00019##

Step 1: Synthesis of (R)-methyl 5-nitro-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (3)

(29) To a solution of methyl 3-hydroxy-5-nitrothiophene-2-carboxylate (1.23 g, 6.05 mmol), (S)-1-(2-(trifluoromethyl)phenyl)ethanol (1 g, 5.26 mmol, 80.00 μ L) in DCM (20 mL) was added PPh.sub.3 (2.07 g, 7.89 mmol), DTBAD (1.82 g, 7.89 mmol) at 25° C. The mixture was stirred at 25° C. for 12 hr. TLC (petroleum ether: ethyl acetate=3:1; R_f=0.75) showed the (S)-1-(2-(trifluoromethyl)phenyl)ethanol was consumed completely and new spots were formed. The mixture was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (40 g SepaFlash® Silica Flash Column, Eluent of 0~5% Ethyl acetate/Petroleum ether gradient @ 30 mL/min) to afford (R)-methyl 5-nitro-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (1.8 g, 4.51 mmol, 85.73% yield, 94% purity) as yellow oil.

Step 2: Synthesis of (R)-methyl 5-amino-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (4)

(30) To a solution of (R)-methyl 5-nitro-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (1.8 g, 4.80 mmol) in EtOH (15 mL) and H₂O (15 mL) were added Fe (1.61 g, 28.78 mmol) and NH₄Cl (1.54 g, 28.78 mmol) at 25° C., the mixture was stirred at 80° C. for 1 hr. TLC (petroleum ether: ethyl acetate=3:1; R_f=0.4) showed the starting material was consumed completely and new spot was formed. The mixture was filtered through a celite pad and the filtrate was concentrated to remove the organic solvent. The solution was diluted with water (50 mL) and extracted with ethyl acetate (20 mL×3). The combined organic phase was washed with brine (20 mL×3), dried over Na₂SO₄, filtered. The filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (12 g SepaFlash® Silica Flash Column, Eluent of 0~30% Ethyl acetate/Petroleum ether gradient @ 75 mL/min) to afford (R)-methyl 5-amino-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (1.4 g, 3.81 mmol, 79.46% yield, 94% purity) was obtained as brown oil which was used for the next step directly. MS(M+H).sup.+ = 345.8

Step 3: Synthesis of (R)-tert-butyl 4-(3-((5-(methoxycarbonyl)-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)amino)-4-nitrobenzyl)piperazine-1-carboxylate (6)

(31) To a solution of (R)-methyl 5-amino-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylate (530 mg, 1.53 mmol), tert-butyl 4-(3-bromo-4-nitrobenzyl)piperazine-1-carboxylate (675.73 mg, 1.69 mmol) and K₂CO₃ (636.33 mg, 4.60 mmol) in t-BuOH (10 mL) were added and Pd₂(dba)₃ (28.11 mg, 30.69 μmol) and XPhos (32.19 mg, 67.53 μmol) under N₂ at 25° C. and the resulting mixture was stirred at 60° C. for 12 h. LCMS showed 37% peak of tert-butyl 4-(3-bromo-4-nitrobenzyl)piperazine-1-carboxylate. Additional Pd₂(dba)₃ (28.11 mg, 30.69 μmol) and XPhos (32.19 mg, 67.53 μmol) were added under N₂ at 25° C. The reaction mixture was stirred at 70° C. for another 12 hrs. The mixture was diluted with ethyl acetate (50 mL) filtered through a celite pad and the filtrate was concentrated. The residue was purified by flash silica gel chromatography (10 g SepaFlash® Silica Flash Column, Eluent of 5~50% Ethyl acetate/Petroleum ether gradient @ 60 mL/min) followed by prep-HPLC (column: Waters Xbridge C₁₈ 150*50 mm*10 μm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 65%-95%, min) and the eluent was lyophilized to afford (R)-tert-butyl 4-(3-((5-(methoxycarbonyl)-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)amino)-4-nitrobenzyl)piperazine-1-carboxylate (400 mg, 601.78 μmol, 39.21% yield) as brown solid. The crude product was used for the next step directly. MS(M+H).sup.+ = 665.1

(32) .sup.1H NMR (400 MHz, CDCl₃) δ=9.71 (s, 1H), 8.16 (br d, J=8.6 Hz, 1H), 7.93 (d, J=7.9 Hz, 1H), 7.68-7.56 (m, 2H), 7.41 (s, 1H), 7.20-7.16 (m, 1H), 6.93 (br d, J=8.4 Hz, 1H), 6.43 (s, 1H), 5.74 (q, J=5.9 Hz, 1H), 3.88 (s, 3H), 3.44 (br s, 6H), 2.38 (br s, 2H), 2.36 (s, 2H), 1.73 (d, J=6.3 Hz, 3H), 1.47 (s, 9H).

Step 4: Synthesis of (R)-tert-butyl 4-((1-(5-(methoxycarbonyl)-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazine-1-carboxylate (7)

(33) To a solution of (R)-tert-butyl 4-(3-((5-(methoxycarbonyl)-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)amino)-4-nitrobenzyl)piperazine-1-carboxylate (0.4 g, 601.78 μmol) in MeOH (20 mL) were added trimethoxymethane (1.60 g, 15.04 mmol, 1.65 mL), formic acid (249.28 mg, 5.42 mmol, 204.32 μL) and Zn (196.75 mg, 3.01 mmol). The reaction mixture was stirred at 70° C. for 12 h. LCMS showed the starting material was consumed completely and one main peak with desired mass. The mixture was filtered and the filter cake was washed with MeOH (50 mL, 50° C.). The filtrate was concentrated under reduced pressure. The residue was diluted with H₂O (15 mL) and adjusted the pH=8 with NaHCO₃ solution. Then the mixture was extracted with EtOAc (20 mL×3). The combined organic layer was washed with H₂O (10 mL×2), dried over Na₂SO₄, filtered. The filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (12 g SepaFlash® Silica Flash Column, Eluent of 0~50% Ethyl acetate/Petroleum ether gradient @ 50 mL/min) to

afford (R)-tert-butyl 4-((1-(5-(methoxycarbonyl)-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazine-1-carboxylate (300 mg, 456.02 μ mol, 75.78% yield, 98% purity) as yellow oil, which was used for the next step directly. MS(M+H).sup.+ = 645.2

Step 5: Synthesis of (R)-5-(6-((4-(tert-butoxycarbonyl)piperazin-1-yl)methyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylic acid (8)

(34) To a solution of (R)-tert-butyl 4-((1-(5-(methoxycarbonyl)-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazine-1-carboxylate (300 mg, 465.33 μ mol) in THF (5 mL) was added a solution of LiOH (33.43 mg, 1.40 mmol) in H.sub.2O (5 mL) drop-wise at 25° C. The mixture was stirred at 40° C. for 12 hr. LCMS showed the starting material was consumed completely and one main peak with desired mass. The reaction mixture was concentrated under reduced pressure to afford (R)-5-(6-((4-(tert-butoxycarbonyl)piperazin-1-yl)methyl)-1H-benzo[d]imidazole-1-yl)-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylic acid (296 mg, crude) as a brown solid, which was used for the next step directly. MS (M+H).sup.+ = 631.0

Step 6: Synthesis of (R)-tert-butyl 4-((1-(5-carbamoyl-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazine-1-carboxylate (9)

(35) To solution of (R)-5-(6-((4-(tert-butoxycarbonyl)piperazin-1-yl)methyl)-1H-benzo[d]imidazole-1-yl)-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxylic acid (296 mg, 464.96 μ mol), NH.sub.4Cl (124.36 mg, 2.32 mmol) in DMF (3 mL) were added HATU (353.59 mg, 929.93 μ mol) and DIPEA (180.28 mg, 1.39 mmol, 242.96 μ L) at 30° C. The mixture was stirred at 30° C. for 2 hr. LCMS showed the starting material was consumed completely and one main peak with desired mass. The mixture solution was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (4 g SepaFlash® Silica Flash Column, Eluent of 0~100% Ethyl acetate/Petroleum ether gradient @ 60 mL/min; Eluent of 0~50% Methanol/Ethyl acetate @60 mL/min) to afford (R)-tert-butyl 4-((1-(5-carbamoyl-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazine-1-carboxylate (400 mg, crude) as brown solid, which was used for the next step directly. MS (M+H).sup.+ = 630.0

Step 7: Synthesis of (R)-5-(6-(piperazin-1-ylmethyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxamide (10)

(36) To a solution of (R)-tert-butyl 4-((1-(5-carbamoyl-4-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazine-1-carboxylate (400 mg, 635.23 μ mol) in DCM (5 mL) was added TFA (1.54 g, 13.51 mmol, 1 mL) and the resulting mixture was stirred at 30° C. for 0.5 hr. LCMS showed the starting material was consumed completely and one main peak with desired mass. The mixture was concentrated under reduced pressure at 10° C. The residue was dissolved with deionized water (40 mL) and lyophilized to afford (R)-5-(6-(piperazin-1-ylmethyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxamide (0.5 g, crude, TFA salt) as a brown solid, which was used for the next step directly. MS(M+H).sup.+ = 530.0

Step 8: Synthesis of (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-((1-(5-carbamoyl-4-((R)-1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazin-1-yl)-4-oxo-6,9,12-trioxa-3-azatetradecan-1-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Compound 2)

(37) To a solution of (R)-5-(6-(piperazin-1-ylmethyl)-1H-benzo[d]imidazol-1-yl)-3-(1-(2-(trifluoromethyl)phenyl)ethoxy)thiophene-2-carboxamide (400 mg, 621.50 μ mol, TFA salt) and (S)-13-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-14,14-dimethyl-11-oxo-3,6,9-trioxa-12-azapentadecyl 4-methylbenzenesulfonate (481.63 mg, 621.50 μ mol) in dioxane (10 mL) were added DIPEA (240.97 mg, 1.86 mmol, 324.76 μ L) and NaI (18.63 mg, 124.30 μ mol, 0.2 eq) at 25° C. The reaction mixture was heated to 100° C. for 12

hours. LCMS showed reagent was consumed completely and one main peak with desired mass. The mixture was concentrated under reduced pressure. The crude product was purified by prep-HPLC (column: Waters Xbridge C.sub.18 150*50 mm*10 um; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 33%-63%, 11 min) and the eluent was lyophilized to afford (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-((1-(5-carbamoyl-4-((R)-1-(2-(trifluoromethyl)phenyl)ethoxy)thiophen-2-yl)-1H-benzo[d]imidazol-6-yl)methyl)piperazin-1-yl)-4-oxo-6,9, 12-trioxa-3-azatetradecan-1-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (48.9 mg, 41.46 μmol, 6.67% yield, 96% purity) as a white solid. MS(M+H).sup.+ = 1132.2 (38).sup.1H NMR (400 MHz, DMSO-d.sub.6) δ=8.97-8.96 (m, 1H), 8.59 (br t, J=6.0 Hz, 1H), 8.49 (s, 1H), 7.94 (d, J=7.8 Hz, 1H), 7.85 (br s, 1H), 7.82-7.74 (m, 2H), 7.68 (d, J=8.3 Hz, 1H), 7.61-7.52 (m, 1H), 7.45-7.35 (m, 5H), 7.33 (s, 1H), 7.28 (dd, J=1.2, 8.3 Hz, 1H), 7.22-7.11 (m, 1H), 7.06 (s, 1H), 5.96-5.92 (m, 1H), 5.15 (d, J=3.5 Hz, 1H), 4.55 (d, J=9.5 Hz, 1H), 4.46-4.41 (m, 1H), 4.40-4.34 (m, 1H), 4.29-4.19 (m, 1H), 3.95 (s, 2H), 3.73-3.39 (m, 16H), 2.44 (br s, 1H), 2.43 (s, 3H), 2.42-2.22 (m, 8H), 2.11-2.00 (m, 1H), 1.95-1.85 (m, 1H), 1.75 (d, J=6.2 Hz, 3H), 0.93-0.91 (m, 9H).

EXPERIMENTAL EXAMPLES

(39) 1. Culture of HeLa Cell Line

(40) The HeLa cell line was purchased from Korea Cell Line Bank (KCLB), Seoul, Korea. The passage in cell culture was maintained at P115 to P125.

(41) For cell counting, cell counter (Thermo Fisher Scientific Inc., Catalog #AMQAX1000) and 0.4% trypan blue solution were used.

(42) For cell culture, DMEM (Gibco, Cat. No. 1195-65; Lot. No. 2085318), FBS (Gibco, Cat. No. 16000-044; Lot. No. 2097593), Penicillin/Streptomycin (PS) (Gibco, Cat. No. 15140-122; Lot. No. 2058855), 100 mm.sup.2 cell culture dish (SPL, Cat. No. 20100), 150 mm.sup.2 cell culture dish (SPL, Cat. No. 20150), 12-well culture plate (SPL, Cat. No. 30012), PBS pH 7.4 (Gibco, Cat. No. 10010-023; Lot. No. 2085080), TrypLE™ Express (Gibco, Cat. No. 12605-010; Lot No. 2070638), Counting Chamber (Hematocytometer) (Hirschmann, Cat. No. 8100204), and 0.4% Trypan Blue Solution (DYNEBIO, Cat. No. CBT3710; Lot. No. 20190723) were used.

(43) 2. Treatment of Compounds of the Present Invention

(44) 2×10⁵ cells were seeded for each well of a 12-well plate (SPL), and the cells were cultured in the culture medium in a total volume of 2 ml.

(45) The compounds of were completely dissolved in DMSO and used in the experiment, and thymidine was completely dissolved in DW and used in the experiment. For thymidine block, the products were treated with 2 mM of thymidine (Sigma-Aldrich Cat. No. T9250-5G) and then incubated for 24 hours.

(46) For release and chemical treatment, the medium was suctioned and washed 3 times with 1×PBS. Complete media was added, followed by incubation for 4 hours in a CO.sub.2 incubator. Each compound was treated according to the concentration of 100 nM and then incubated for 6 hours again.

(47) 3. Western Blotting

(48) For SDS-PAGE and Western blotting, 1×RIPA lysis buffer (Rockland, Cat. No. MB-030-0050; Lot no. 39751), 100×Protease Inhibitor Cocktail (Quartett, Cat. No. PPI1015; Lot no. PC050038424), Pierce custom character BCA protein assay kit (ThermoScientific, Cat. No. 23225; Lot no. UC276876), albumin standard (ThermoScientific, Cat. No. 23209; Lot no. UB269561), 4-15% Mini-PROTEAN TGX stain-free gel (Bio-rad, Cat. No. 4568085; Lot no. L007041B), 10×Tris/Glycine/SDS buffer (Bio-rad, Cat. No. 1610732; Lot no. 10000044375B); 10×TBS (Bio-rad, Cat. No. 1706435; Lot no. 1000045140B), 10% Tween 20 (Cat. No. 1610781; Lot no. L004152B), Color protein standard broad range (NEB, Cat. No. P7719S; Lot no. 10040349), 4× Laemmli sample buffer (Bio-rad, Cat. No. 1610747; Lot no. L004133B), β-mercaptoethanol (Sigma-Aldrich, Cat. No. M3148; Lot no. 60-24-2), SuperBloc custom character

T20 (TBS) blocking buffer (ThermoScientific, Cat. No. 37536; Lot no. UC282578), 1M sodium azide solution (Sigma-Aldrich, Cat. No. 08591-1 mL-F; Lot no. BCBV4989), aI-Rabbit pAb to Ms IgG (abcam, Cat. No. ab97046; Lot no. GR3252115-1), aGoat pAb to Rb IgG (CST, Cat. No. 7074S; Lot no. 28), α -GAPDH (abcam, Cat. No. ab8245; Lot no. GR3275542-2), α -Plkl (CST, Cat. No. 208G4), α -BRD4 (CST, Cat. No. 13440S), ECL⁺ custom character Prime western blotting reagents (GE Healthcare, Cat. No. RPN2232; Lot no. 17001655), Ponceau S solution (Sigma-Aldrich, Cat. No. P7170; Lot no. SLBV4112), Difco⁺ custom character Skim milk (BD, Cat. No. 232100; Lot no. 8346795), and iBlot[®] 2 NC Regular stacks (Invitrogen, Cat. No. IB23001; Lot no. 2NR110619-02) were used.

(49) For cell harvesting, the cells were first separated from the plate using trypsin and then washed with the medium and PBS. Specifically, the medium was suctioned off and washed with 1 mL of PBS, and PBS was suctioned off. The cells were treated with 0.5 mL TrypLE⁺ custom character Express at 37° C. for 7 minutes to separate the cells, and then 0.5 mL of complete medium was added to collect 1 mL of cell culture solution. Then, 1 mL of the cell collection solution was centrifuged at 8,000 rpm for 120 seconds, and the supernatant was removed. After washing with 0.2 mL of PBS, the PBS was removed.

(50) For cell lysis, a lysis buffer was added and cell debris was removed to obtain a cell lysate. Specifically, the cells were treated with 70 μ L of 1 $\times$ RIPA buffer containing a protease inhibitor and incubated for 30 minutes on ice. Then, the cells were centrifuged at 4° C. and 15,000 rpm for 10 minutes to obtain a cell lysate.

(51) Then, a standard curve was obtained using the BCA assay, and the protein mass in the lysate was quantified by substituting the curve equation. The mixture was incubated at 37° C. for 30 minutes using 20 μ L of standard or sample solution, and 200 μ L of BCA or Bradford solution, and measured at 562 nm absorbance. Samples were prepared by adding 4 $\times$ sample buffer so that the quantity of protein added to each well was 15 μ g.

(52) Sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) was performed by setting a running time of 100 minutes at 120 V on a 4-15% Mini-PROTEAN TGX stain-free gel (15 well). Transferring was performed on iBlot[®] 2 NC Mini stacks at P0 mode of the dry blotting system. After staining using Ponceau S solution, blocking was performed for 1 hour with a blocking buffer (Thermo). After washing with 1 $\times$ TBS containing 0.05% Tween 20, the product was reacted at 4° C. for 16 hours with anti-Plkl (CST) antibody (1:500), anti-BRD4 (Cell signaling) antibody (1:1000) or anti-GAPDH (abcam) antibody (1:10,000) in 1 $\times$ TBS-T as a primary antibody. After washing three times for 10 minutes with 1 $\times$ TBS containing 0.05% Tween20, the product was reacted at room temperature for 1 hour with anti-mouse antibody (abcam) (1:10000) or anti-rabbit antibody (CST) (1:5000) in 1 $\times$ TBS-T as a secondary antibody. Then, after washing three times for 10 minutes with 1 $\times$ TBS containing 0.05% Tween 20, the product was detected with an ECL working solution (1:1).

(53) To analyze the results, an image analyzer (GE) was used to obtain final blot data. The ratio of PLK1 to GAPDH for each sample was calculated using the ImageQuant TL (ver.8.2.0) program. Each calculated value was entered into each cell of the Graphpad Prism 9 program, and the graph was automatically calculated to confirm the Dmax value corresponding to the protein degradation ability.

(54) 4. Confirmation of PLK1 Degradability of the Compounds of the Present Invention

(55) As a result of the experiment, it was confirmed that all the compounds of the examples of the present invention exhibited a PLK1 degradability of 40 to 60%.

(56) The contents of all references, patents, pending patent applications and published patents, cited throughout this application are hereby expressly incorporated by reference.

(57) Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the disclosure described herein. Such equivalents are intended to be encompassed by the following claims. It is understood that the

detailed examples and embodiments described herein are given by way of example for illustrative purposes only, and are in no way considered to be limiting to the disclosure. Various modifications or changes in light thereof will be suggested to persons skilled in the art and are included within the spirit and purview of this application and are considered within the scope of the appended claims. For example, the relative quantities of the ingredients may be varied to optimize the desired effects, additional ingredients may be added, and/or similar ingredients may be substituted for one or more of the ingredients described. Additional advantageous features and functionalities associated with the systems, methods, and processes of the present disclosure will be apparent from the appended claims. Moreover, those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the disclosure described herein. Such equivalents are intended to be encompassed by the following claims.

Claims

1. A compound represented by the following Formula I, or a stereoisomer thereof, or a pharmaceutically acceptable salt thereof:
ULM-Linker-PTM [Formula I] wherein in the Formula I above, ULM is E3 ubiquitin ligase binding moiety represented by the following Formula A or B: ##STR00020## wherein X.sub.1 is —CH.sub.2— or —CO—; ##STR00021## wherein ##STR00022## is a 5-membered heteroaryl; and Y.sub.1 is hydrogen or C.sub.1-3 alkyl; PTM is PLK1 binding moiety represented by the following Formula II: ##STR00023## and Linker is a chemical group that links ULM and PTM.
 2. The compound of claim 1, wherein ULM is represented by Formula A and Formula II is represented by the following formula: ##STR00024##
 3. The compound of claim 1, wherein ULM is represented by Formula B and Formula II is represented by the following formula: ##STR00025##
 4. The compound of claim 1, wherein the compound is compound 1 or compound 2
##STR00026##
 5. The compound of claim 1, wherein the compound is a bifunctional compound that induces PLK1 protein degradation.
 6. A composition for prevention or treatment of a PLK1 related disease, the composition comprising the compound according to claim 1, or a stereoisomer thereof, or a pharmaceutically acceptable salt thereof.
 7. The composition of claim 6, wherein the PLK1 related disease is a cancer, a benign tumor, or a neurological disease.
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